

Matched Dale-constrained COBA and PING learn a small SHD task

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Abstract

Both matched Dale-constrained conductance-based networks learn the small Spiking Heidelberg Digits (SHD) task above the registered threshold. COBA reaches a best held-out accuracy of 31%, and PING reaches 34.7%, against the 5% chance floor and the pre-registered 20% success criterion. Neither run records a skipped update or non-finite forward batch. Fixed test utterances replayed through the trained input and recurrent weights produce directly comparable input, excitatory, and inhibitory rasters. This one-seed result establishes feasibility; it does not compare the architectures' expected accuracy or efficiency.

Methods

SHD contains event-based recordings of spoken digits across 700 cochlear channels and 20 classes. The registered protocol has three ordered stages:

1. **Gate with a local smoke test.** Train seed 42 on 128 training utterances for two epochs. Require finite loss, active hidden populations, saved checkpoints, and no skipped updates before allocating cloud compute. The original shared input-weight mean passes, so the single allowed shared adjustment is not used.
2. **Train the pilot.** Train each cell for 20 epochs on the same 1000-utterance subset. Both cells have 256 excitatory and 64 inhibitory cells, Dale's law, fixed recurrence, 95% input sparsity, input-weight mean 0.9, membrane-mean readout scaled by 225, Adam learning rate 0.0004, batch size 32, integration timestep 1 ms, and duration 1000 ms. COBA disables the inhibitory loop and uses voltage-gradient dampening 1 PING enables the fixed loop and uses dampening 1000.
3. **Record matched dynamics.** Select official test positions 0, 1, 2 before inspecting predictions, bin their cochlear events on the same integration grid, and replay the identical spike tensors through both trained input and recurrent weight sets. The replay omits the classifier output tensor because the batch-raster interface constructs a ten-output probe head; the omitted readout does not feed back into the hidden E/I dynamics plotted here.

Every published raw file is listed with its SHA-256 digest in `raw_sha256.json`. The artifact manifest and reproducer record the publishing commit and exact runner invocation.

Results

Both cells cross the registered accuracy criterion

COBA peaks at 31% in epoch 19 and ends at 28.4%. PING peaks at 34.7% in epoch 9 and ends at 31.7%. Both exceed the registered criterion, so the hypothesis survives this feasibility pilot.

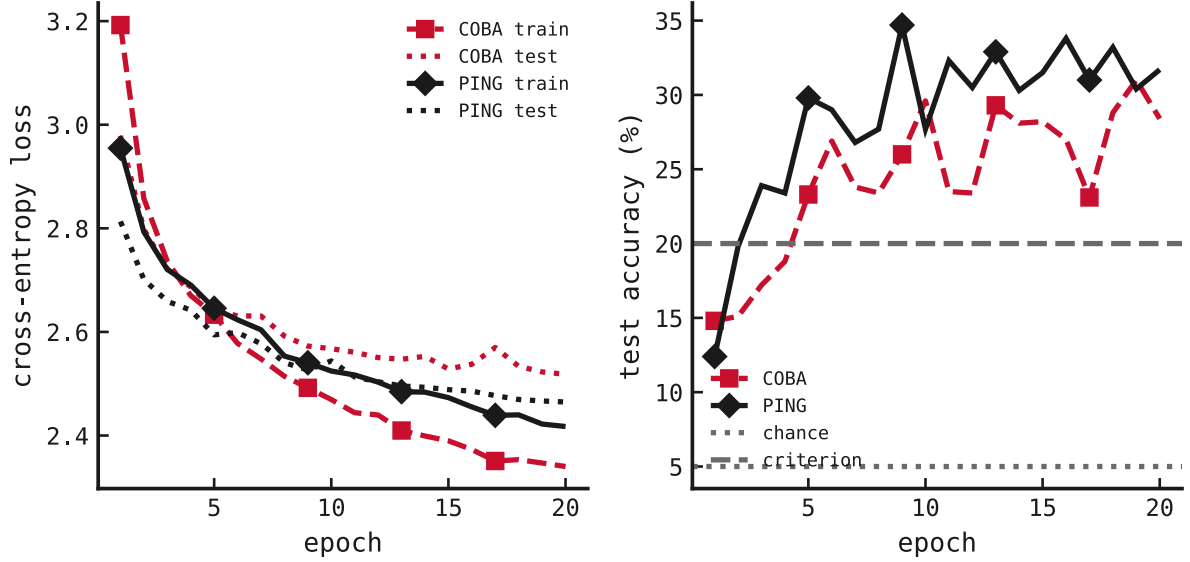


Figure 1: Matched learning curves for the registered SHD pilot. The left panel shows cross-entropy loss against epoch for training and held-out test subsets; the right panel shows held-out classification accuracy in percent. Red dashed traces and square markers identify COBA; black solid traces and diamond markers identify PING, with line style and marker shape retaining the distinction in grayscale. Horizontal lines mark 5% chance and the 20% registered criterion. COBA reaches 31% and PING 34.7%; both train above the criterion.

Activity stays finite and architecture-appropriate

At the final training epoch, COBA's excitatory population fires at 21.5 Hz and its disabled inhibitory population remains at 0 Hz. PING ends at 10.3 Hz excitatory and 29 Hz inhibitory activity. Neither cell is silent or saturated.

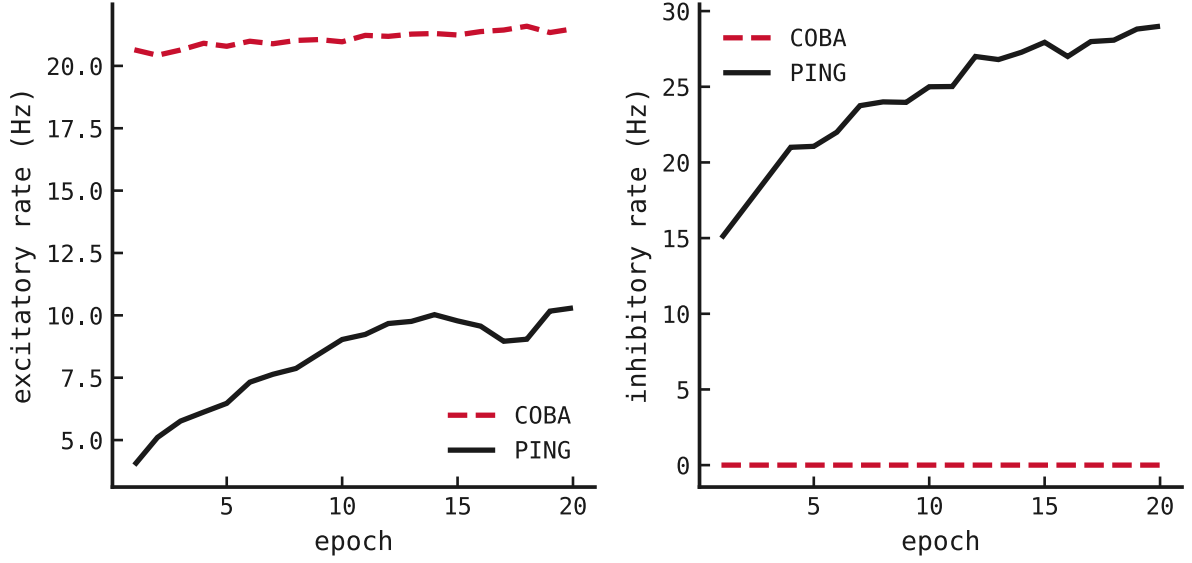


Figure 2: Hidden-population firing rates during training. The horizontal axis is epoch; the vertical axes are excitatory rate in hertz (left) and inhibitory rate in hertz (right). Red dashed traces denote COBA and black solid traces PING. COBA’s disabled loop gives zero inhibitory activity while its excitatory population remains active. PING sustains both populations without a non-finite batch or skipped update.

Identical utterances expose comparable E/I dynamics

The matched replay gives COBA 27.1 Hz excitatory activity and no inhibitory activity. PING gives 7.8 Hz excitatory and 32.7 Hz inhibitory activity. The shared input rows are identical within each test-position column by construction.

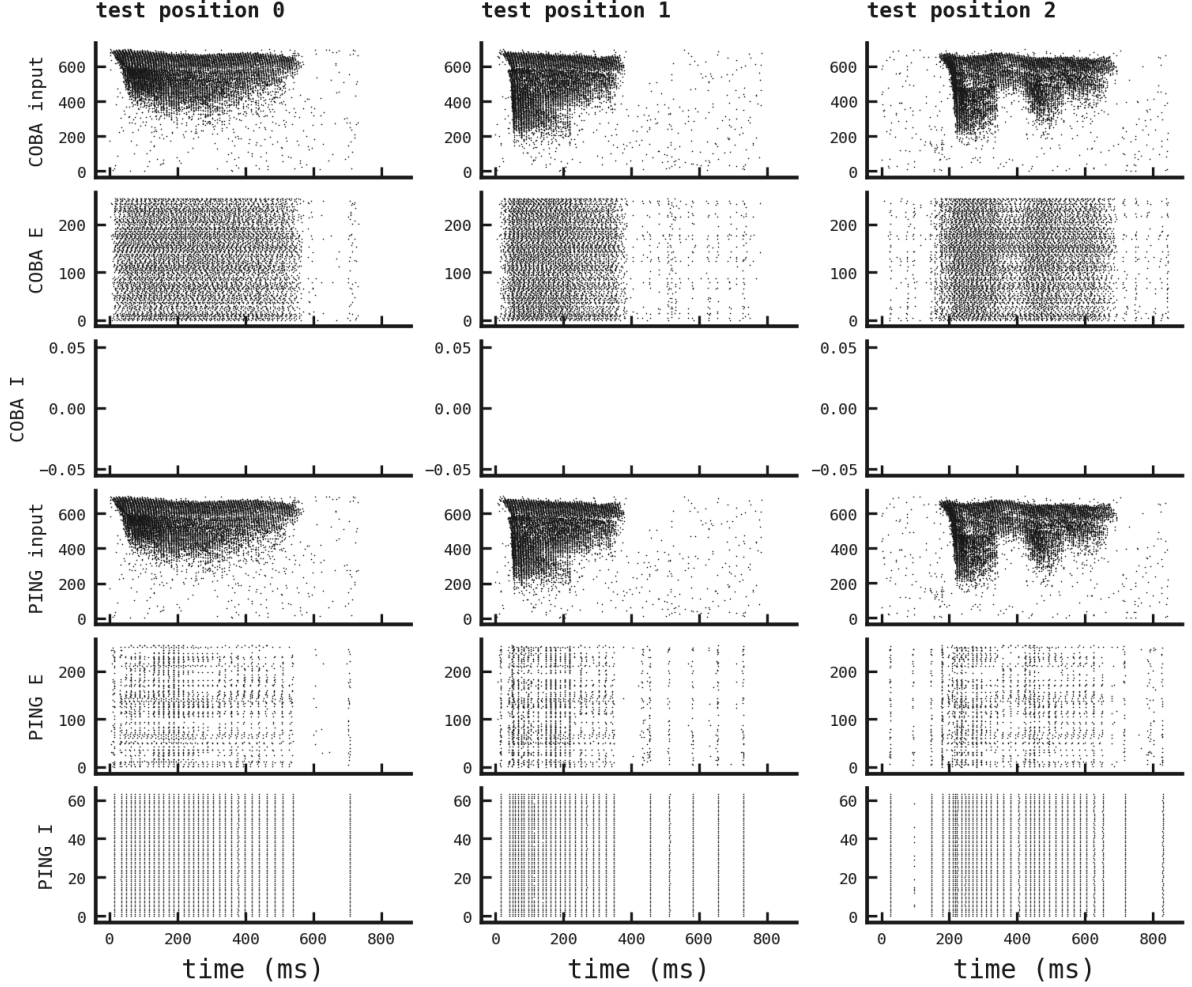


Figure 3: Matched spike rasters for the preselected official SHD test positions 0, 1, 2. Columns are utterances and share a time axis in milliseconds. Within each architecture, rows show the 700-channel cochlear input, 256 excitatory cells (E), and 64 inhibitory cells (I), with fixed population order. COBA and PING receive byte-identical binned input spikes. COBA has active E cells and a silent disabled I loop; PING produces active E and I populations on the same utterances.

Verdict and limits

The registered feasibility criterion is met: both matched networks train above 20% and yield comparable population recordings. This is one pre-registered seed on a small subset. It cannot establish a reliable accuracy difference, a PING advantage, or generalisation to full-scale SHD. Two cloud evidence-capture attempts failed after valid training because the single-sample replay path could not emit SHD snapshots; the final rasters use the existing arbitrary-input batch interface locally. Total RunPod spend is 0.42 USD, and 0 pods remain active.